Kepler's Third Law in Multiple Systems

Component Orbital Parameters and Controlled Group Approximations

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Abstract

This study develops the operational interpretation of Kepler's third law through $E = \omega^2 R^3$ for circular orbits and $E = n^2 a^3$ for relative Keplerian ellipses. The approach applies to individual celestial sources with their own gravitational flows and to systems containing several such sources. It distinguishes a source from a body moving in another source's field. Trajectories, field-error bounds and the equilateral stability criterion are expressed through orbital parameters, which can be inferred from suitable orbital lengths and periods or prescribed for synthetic calculations. For an isolated binary, the relative Kepler parameter is exactly $E_1 + E_2$. For a spatial group, replacing its members by one centre is an approximation whose error depends on geometry. We derive a conservative exterior-field bound and distinguish external tidal contamination from internal monopole error. Nine prescribed two- and three-body configurations cover equal and unequal binaries, four positions of an outer companion, inclined configurations and an equilateral benchmark. An initial pair-plus-third configuration at separation 20 admits a sufficient radial interval 4.7353–6.0788 in which each of two separately defined errors is at most 5%. Closer configurations need not admit such an interval. Calculations for Kepler-16, KOI-126 and TIC 454140642 provide observational comparisons and hierarchy tests. In the quadruple, the largest sampled error of the leading tidal contrast is 2.595% relative to the exact external contrast, while the corresponding radial-velocity reconstruction retains substantial residuals. Numerical approximation accuracy and observational agreement therefore require separate assessment. For the positive-source configurations examined here, the adopted equations have the mathematical form studied in the three- and N-body problem. This permits established coordinate constructions, approximation methods and stability criteria to be used with their assumptions retained. The contribution is a reproducible calculation of specified configurations and controlled group approximations through orbital parameters. It is not a new general analytical solution or an observational demonstration of the proposed flow mechanism.

Keywords: Kepler's third law, orbital parameter, gravitational flow, three-body problem, binary stars, hierarchical systems, circumbinary planets, error bounds.

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1 Introduction and position in the series

The proposed non-material medium, to which the author attributes properties of an ideal incompressible fluid or gas with zero internal viscosity, is called the E-field in the companion study [4].

Main results. The calculation recovers individual binary source parameters from suitable orbital information, derives explicit error domains for a summed group parameter, and quantifies the leading differential contribution between two pairs. Synthetic configurations and published systems provide separate numerical and observational comparisons. A predeclared diagnostic also evaluates direct outer-to-inner period ratios for both pairs of one quadruple.

The first two Ew2R3 preprints develop the orbital parameter E as an operational quantity inferred from trajectories [1, 2]. The third examines a flow interpretation [3]. In the worldview-oriented form of Kepler's third law, the parameter characterizes each celestial source with its own gravitational flow, rather than being restricted to multiple-star systems. For a central source, a suitable orbit around that source provides its parameter. A body's orbit around a different source does not, by itself, determine the orbiting body's own source parameter.

The present study extends the calculation from one source and an orbiting body to a comparable binary, triples, a hierarchical quadruple and a circumbinary planet. Flow generation is not assigned to every material body. An object acting as a source and an object responding to an external field have distinct roles. Flow transition and body–flow coupling are separate physical questions, examined in the companion studies [5, 6]. The author's broader matter–medium hypothesis is presented separately [4].

The calculation uses an explicitly stated rule for combining the fields of the selected sources. Kepler's third-law relation alone does not specify this multi-source rule. For the positive-source configurations considered here, the resulting equations overlap mathematically with the three- and N-body problem. This overlap makes established mathematical results applicable without identifying their physical interpretation with the proposed flow interpretation.

We distinguish an exact binary reduction, an error-controlled exterior approximation, and the motion of a subgroup perturbed by an external companion. A sum of parameters does not create a sharp physical boundary around the sources. "Inside the common region" means inside a specified approximation contour, rather than inside an observed surface of a medium. Coordinates are inertial unless stated otherwise.

The companion study of external flow around impermeable atoms [17] examines pressure-gradient response and conditional added inertia. Its multi-source potential is prescribed rather than derived from atomic geometry. The orbital calculations here use the inverse-square source equations and do not include an additional relative-velocity response.

2 Related work and scope of contribution

Circumbinary planetary dynamics and its non-Keplerian oscillations have established analytic and numerical treatments, including Leung and Lee [9]. Secular dynamics of hierarchical 2+2 and $3+1$ quadruples is studied by Hamers and Lai [10]. Kepler-16, KOI-126 and TIC 454140642 are previously observed systems [11, 12, 13, 14]. Their discovery and their dynamical architectures are not new results of the present work.

The contribution sought here is a common reporting procedure: an explicit group centre, a conservative field-error bound, an external-perturbation bound and a separate check against published observables. No claim of priority is made for Kepler scaling, multipole expansions

or tidal linearization. Expressing the equations in orbital parameters does not change their mathematical content. Likewise, agreement between two integrations of the same equations is a numerical check, not an independent test of a proposed medium.

3 Source parameters and equations of motion

3.1 Selected sources and the adopted field prescription

Let $E_j > 0$ characterize a selected source with its own gravitational flow. The parameter has dimensions L^3T^{-2}. Source status is not assumed for every material object. A body treated as a receiver without its own source contribution has $E_j = 0$ in this calculation. An unknown source parameter must instead remain undetermined, because a lack of suitable orbital data does not establish a zero value. All stellar sources in the reported benchmarks have positive parameters.

For circular motion around an isolated source, $g = \omega^2 R$ and $E = \omega^2 R^3$ give $g = E/R^2$. Extending this radial contribution to a moving source defines the model field $\mathbf{g}_j(\mathbf{x}) = E_j(\mathbf{r}_j - \mathbf{x})/|\mathbf{r}_j - \mathbf{x}|^3$. The calculation assumes that the vector contributions of the selected sources add. This superposition is an additional modelling assumption, not a derivation from Kepler's third law alone and not a premise that every object produces a gravitational flow. The equations of motion are

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \qquad \dot{\mathbf{v}}_i = \sum_{j \neq i} E_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}. \tag{1}$$

Here $\dot{\mathbf{v}}_i$ is the rate of change of the body's velocity, calculated from the source fields. "Equation of motion" denotes this explicit prescription rather than a separately named physical law. The field-to-body relation has not been derived here from a velocity field $\mathbf{u}$. The prescription applies away from collisions and does not describe stellar interiors, dissipative tides, relativity or stellar deformation.

For some published systems, the adopted E_i are rescaled outputs of existing dynamical fits, using a common solar orbital calibration and consistent units for all components. These inputs preserve the assumptions of the cited fits and are not new independent measurements of every source parameter. The synthetic experiments prescribe their E_i directly.

Define $E_G = \sum_{i \in G} E_i$ and $\mathbf{c}_G = \sum E_i \mathbf{r}_i / E_G$. Pairwise antisymmetry gives constant total $\sum E_i \mathbf{v}_i$ for an isolated system. The corresponding conserved quantities include

$$H = \frac{1}{2} \sum_i E_i |\mathbf{v}_i|^2 - \sum_{i<j} \frac{E_i E_j}{|\mathbf{r}_i - \mathbf{r}_j|}, \tag{2}$$

$$\mathbf{L} = \sum_i E_i \mathbf{r}_i \times \mathbf{v}_i. \tag{3}$$

These are computational invariants in the chosen orbital-parameter normalization.

For a binary, $\mathbf{s} = \mathbf{r}_2 - \mathbf{r}_1$ obeys

$$\ddot{\mathbf{s}} = -(E_1 + E_2)\frac{\mathbf{s}}{s^3}, \qquad n^2 a_{\mathrm{rel}}^3 = E_1 + E_2, \qquad n = 2\pi/P. \tag{4}$$

The period relation holds for an ellipse with relative semimajor axis a_{rel}. Eccentricity does not by itself invalidate it. Neither the instantaneous separation nor one component's barycentric radius can be silently substituted for a_{rel}. A finite companion contributes its own E_i.

3.2 Kinematics around one isolated source

Take a receiver whose own source contribution is neglected, moving outside one isolated source with fixed orbital parameter E. With the source at the origin, Eq. (1) reduces to $\ddot{\mathbf{r}} = -E\mathbf{r}/r^3$, where $r = |\mathbf{r}|$. In this central model the specific angular momentum $\mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}}$ is constant. A circular orbit of radius R has tangential speed $v_{\rm circ}$, angular speed ω and period P, so that

$$v_{\rm circ}^2 = \frac{E}{R}, \qquad \omega^2 R^3 = E, \qquad P = 2\pi\sqrt{\frac{R^3}{E}}. \tag{5}$$

For a bound ellipse of semimajor axis $a_{\rm orb}$ and eccentricity e, the same model gives

$$n^2 a_{\rm orb}^3 = E, \qquad n = \frac{2\pi}{P}, \qquad |\dot{\mathbf{r}}|^2 = E\left(\frac{2}{r} - \frac{1}{a_{\rm orb}}\right), \qquad |\mathbf{h}|^2 = E a_{\rm orb}(1-e^2). \tag{6}$$

The speed identity follows from the constant orbital energy per unit receiving-body normalization, $|\dot{\mathbf{r}}|^2/2 - E/r = -E/(2a_{\rm orb})$. The instantaneous r varies along the ellipse and cannot replace $a_{\rm orb}$ in the period relation. These are the one-source kinematics underlying the earlier orbital studies [1, 2]. The velocity here belongs to the receiving body. It is not an identification with a velocity of the proposed medium [3, 5, 6]. When both components have non-negligible source parameters, their relative motion instead involves the sum derived above and used in the next subsection.

3.3 Recovering the individual parameters from binary kinematics

Let a_1 and a_2 be the semimajor axes of the two barycentric orbits, with $a = a_{\rm rel} = a_1 + a_2$. For the isolated binary, the centre condition gives $E_1 a_1 = E_2 a_2$. Combining it with $E_1 + E_2 = n^2 a^3$ yields

$$E_1 = n^2 a^2 a_2, \qquad E_2 = n^2 a^2 a_1. \tag{7}$$

The opposite component's barycentric axis appears in each expression. Using $n^2 a_1^3$ for E_1 would be incorrect. For example, at $(E_1, E_2) = (0.75, 0.25)$ and $a = 1$, one has $(a_1, a_2) = (0.25, 0.75)$ and $n = 1$. Eq. (7) recovers both prescribed parameters. These component values distinguish the source parameters from the relative-orbit parameter.

The input information matters. An orbital period and a relative semimajor axis determine the sum, not the split. A barycentric axis ratio supplies the split. For double-lined spectroscopic semiamplitudes K_1, K_2, eccentricity e and inclination i, the projected axes satisfy $a_j \sin i = K_j\sqrt{1-e^2}/n$. Substitution gives

$$E_1 \sin^3 i = \frac{P}{2\pi}(1-e^2)^{3/2}(K_1+K_2)^2 K_2, \tag{8}$$

$$E_2 \sin^3 i = \frac{P}{2\pi}(1-e^2)^{3/2}(K_1+K_2)^2 K_1. \tag{9}$$

Without the inclination these are projected combinations, not unique E_i. Physical, rather than angular, astrometric axes also require a distance scale. These identities are consequences of the adopted binary model, not new evidence for a flow. Parameter errors and their correlations must be propagated together. Rounded published solutions cannot be assigned zero uncertainty.

4 Spatial error of replacing a group by one centre

Set $\mathbf{s}_i = \mathbf{r}_i - \mathbf{c}_G$, $b = \max_i |\mathbf{s}_i|$ and $m_2 = \sum E_i |\mathbf{s}_i|^2 / E_G$. At $\mathbf{x}$ relative to the centre, $R = |\mathbf{x}| > b$, compare

$$\mathbf{g}_G = \sum_i E_i \frac{\mathbf{s}_i - \mathbf{x}}{|\mathbf{s}_i - \mathbf{x}|^3}, \quad \mathbf{g}_0 = -E_G \frac{\mathbf{x}}{R^3}, \quad \epsilon_G = \frac{|\mathbf{g}_G - \mathbf{g}_0|}{E_G/R^2}. \tag{10}$$

The vanishing weighted dipole and a Taylor remainder give the sufficient bound

$$\epsilon_G \le B_G(R) = \frac{3m_2R^2}{(R-b)^4}. \tag{11}$$

Its derivation is in Appendix A. For fixed positions, B_G decreases from infinity to zero on $R > b$. The root $B_G(R_\varepsilon) = \varepsilon$ defines a conservative exterior sphere. It is sufficient, generally not necessary, and changes as the configuration evolves. It is not a stability radius or the edge of an interaction.

For a pair and an external third source at distance D from the pair centre, work in a translating, nonrotating frame with the actual weighted centre acceleration subtracted. A sufficient bound for the external differential acceleration, normalized to E_G/R^2, is

$$\epsilon_{\text{ext}} \le \frac{E_3}{E_G}\left[\frac{2R^3}{(D-R)^3} + \frac{3m_2R^2}{(D-b)^4}\right], \quad b < R < D. \tag{12}$$

The first term bounds the variation across the trial radius. The second corrects the difference between the acceleration of the centre point and that of the weighted, extended pair. Both errors must be reported. If each is bounded by 5%, their sum is bounded by 10%, not 5%.

5 Two- and three-source configurations

5.1 Initial conditions and numerical protocol

Use pair separation $L_0 = 1$, pair parameter $E_G = 1$ and time unit $T_0 = \sqrt{L_0^3/E_G}$. The reference binary period is $2\pi T_0$. The unequal pair has $(E_1, E_2) = (0.75, 0.25)$. A third source has $E_3 = 0.3$. Jacobi initial conditions place the third source at $(D, 0, 0)$ relative to the pair centre with velocity $\sqrt{1.3/D}(0, \cos i, \sin i)$, followed by the common weighted-centre shift. These are prescribed circular initial estimates, not exact three-body circular solutions. The equal triangle has side one and $E_i = 1/3$.

Fourth-order Runge–Kutta calculations use $h = 2\pi/4096$ and $h/2$. The requested interval is eight inner reference periods, except for the triangle, whose accepted analytic comparison spans one period. Integration stops on a pair separation below 0.15. A stopped trajectory is not extrapolated beyond that event.

Table 1: Initial exterior 5% sufficient radii. Distances are in L_0.

Configuration	D	$R_{5\%,\,\text{all}}$	Third vs pair's initial sphere
Equal binary	–	4.82113	–
Unequal binary	–	4.73531	–
Unequal pair + third	1.25	6.81695	inside
Unequal pair + third	3	14.47047	inside
Unequal pair + third	8	37.57357	outside
Unequal pair + third	20	93.57758	outside
Inclined triple, $i = 60°$	3	14.47047	inside
Inclined triple, $i = 60°$	20	93.57758	outside
Equal equilateral triple	–	5.56696	not a pair hierarchy

Inclination changes the subsequent evolution, although it leaves these initial positions and initial spherical bounds unchanged. Membership of the larger all-source approximation sphere is a geometrical construction and does not imply that the actual field inside that sphere is central.

5.2 Results and limits

The 36 numerical checks passed for their declared intervals and definitions. The maximum common-sample position difference under step halving was $9.64 \times 10^{-12} L_0$, and the maximum relative drift of H was 6.4×10^{-10}. Binary analytic errors were at most $5.69 \times 10^{-13} L_0$. For $D = 1.25$ a close-approach stop occurred at $t \simeq 0.553 T_0$. It is not an eight-period validation. The equal equilateral initial state is the circular Lagrange solution, which is linearly unstable for these parameters (see Section 6). An attempted eight-period comparison departed from the exact solution by $0.938 L_0$, whereas the one-period error was $2.75 \times 10^{-14} L_0$. The longer comparison is retained as an instability-sensitive diagnostic and is not included among the successful long-duration accuracy checks. Growth of round-off or truncation perturbations is consistent with the known instability. This departure alone neither proves an integrator defect nor validates its accuracy. No general stability theorem follows.

An angular/radial scan used 192 radii and 2048 directions, compared with a 96-by-512 scan. For coplanar triples at $D = 1.25, 3, 8, 20$, the best sampled maximum of the separate internal and external errors was approximately 376.86%, 38.79%, 7.68% and 2.15%, respectively. The coarse/fine change reached 34.11 percentage points in the closest case. A sampled absence of a suitable radius is therefore not automatically a proof of its absence in the continuum.

For $D = 20$ the sampled simultaneous 5% interval was approximately 3.7748–6.9565. The stronger statement follows from Eqs. (11) and (12): the initial interval 4.735314–6.078802 is sufficient for each error separately in every direction. These are initial spatial statements. Eight inner periods do not cover even one outer period in this case.

Table 2: Duration actually checked and maximum common-sample position change under step halving. These are synthetic integrations without planets or observations.

Configuration	Duration in $2\pi T_0$	$\max \lvert\Delta \mathbf{r}\rvert / L_0$
Equal binary	8	3.02×10^{-12}
Unequal binary	8	5.24×10^{-12}
Triple, $D = 1.25$, stopped	0.088013	9.64×10^{-12}
Triple, $D = 3$	8	6.74×10^{-12}
Triple, $D = 8$	8	4.38×10^{-12}
Triple, $D = 20$	8	2.00×10^{-12}
Triple, $D = 3$, $i = 60°$	8	5.80×10^{-12}
Triple, $D = 20$, $i = 60°$	8	4.78×10^{-12}
Equal triangle	1	5.11×10^{-13}

Small values in this table describe convergence of the adopted equations over the stated interval. They cannot be substituted for an approximation error, observational residual, or long-term stability test. The planet comparison below belongs to Kepler-16. It does not test planets in every synthetic configuration.

5.3 Configuration coverage and the observer's frame

The preceding nine experiments are a bounded test set, not an enumeration of all multiple-star arrangements. A general initial-condition description uses nested groups: $2+1$, $2+2$, $(2+1)+1$, $2+3$, $4+1$, and larger trees. Each branch has its own separation scale, eccentricity, orbital phase, inclination and ascending node. Non-hierarchical states instead require explicit positions and velocities. Changing the viewing direction alone does not change any of these physical initial conditions.

For a group G, define $\mathbf{R}_G = \sum_{i \in G} E_i \mathbf{r}_i / \sum_{i \in G} E_i$. Then $\mathbf{r}_i = \mathbf{R}_G + \boldsymbol{\rho}_{i,G}$ separates motion of the group from internal motion. A trajectory plotted relative to $\mathbf{R}_G(t)$ must subtract its value

at every historical time, not only its present value. Keplerian ellipses may initialize each branch. The subsequent calculation must include all pair interactions rather than constrain the bodies to those ellipses.

The Ew2R3 visualization project has an archived software release [7]. The atlas implementation used in this study extends the configuration handling to two to twelve bodies, including ten presets and direct Cartesian input. The archived release and this atlas extension are distinct versions. The implementation uses Eq. (1), including its explicit assumption of vector superposition. A scalar orbital invariant alone does not specify the multi-source equations.

A short implementation check compares 128 steps with 256 half-steps for each preset. The largest position difference is $5.63 \times 10^{-6} L_0$. Pairwise distances under changes of the displayed frame differ by at most $1.43 \times 10^{-14} L_0$. These results check the construction and coordinate handling over the stated short intervals. They do not constitute an observational comparison or a long-term stability result.

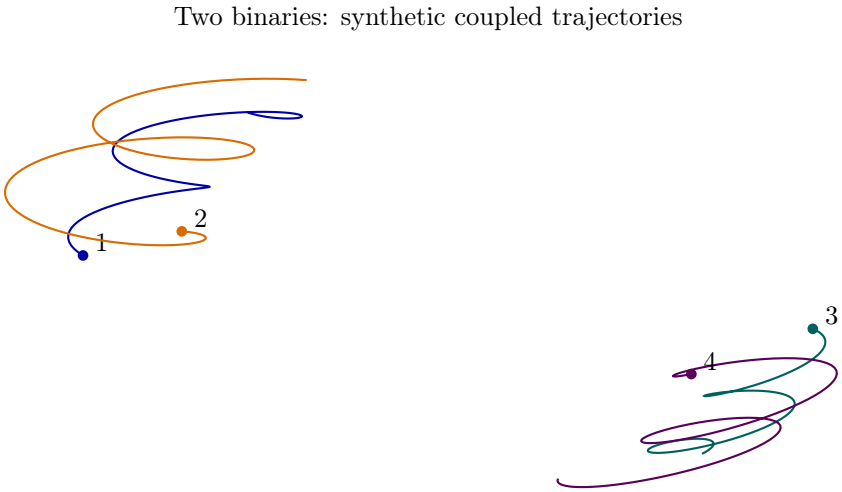


Figure 1: Synthetic $2 + 2$ example with distinct orbital planes. All four sources are integrated together. Ellipses initialize the state and do not constrain subsequent motion. The fixed projection has independently scaled horizontal and vertical plotting axes, so apparent angles are not physical orbital inclinations. This is a numerical illustration, not an observed system.

6 Mathematical overlap with the three- and N-body problem

6.1 Scope of the mathematical correspondence

Equation (1) has the mathematical form of the point-body equations studied in celestial mechanics as the three-body problem for $N = 3$ and the N-body problem more generally. Results depending on those equations can therefore be expressed in the parameters E_i, with their original assumptions retained. This acknowledges a mathematical overlap, not a transfer of physical interpretation. The present article did not originate as a solution of the general three-body problem and does not supply a new general analytic solution.

6.2 Hierarchies, multipoles and differential perturbations

The pair-plus-third construction uses Jacobi coordinates: the inner relative coordinate and the third source relative to the inner weighted centre. These are hierarchical triple coordinates, not a new architecture. The same distinction between internal and group motion applies to the observed triples and the $2 + 2$ hierarchy, whose architectures predate this work [13, 14, 10].

The cancellation of the weighted dipole in Appendix A is the standard reason that the leading exterior correction is quadrupolar. Equation (11) has the large-distance form

$$B_G(R) = \frac{3m_2}{R^2}\left(1 - \frac{b}{R}\right)^{-4} = \frac{3m_2}{R^2}\left[1 + O\left(\frac{b}{R}\right)\right]. \tag{13}$$

It thus has the usual second-order relative spatial scaling, but bounds the full remainder for $R > b$ rather than merely retaining a leading term. It is conservative and loses usefulness near $R = b$. No superiority over existing bounds is claimed. The reporting choice is to keep this all-direction sufficient bound separate from external contamination and from observed residuals.

For $b \ll R \ll D$, the leading part of Eq. (12) scales as $2(E_3/E_G)(R/D)^3$, the same differential scale derived in Eq. (18). This motivates comparison with a Hill scale, but not identification of the two. In a circular restricted hierarchy with a compact group and $E_G/E_3 \ll 1$, balancing the internal scale against the rotating-frame linear differential term defines $r_H = D[E_G/(3E_3)]^{1/3}$. Our translating, nonrotating field bound uses different terms and asks for a specified fractional error. It is neither a rotating-frame equilibrium criterion nor a long-term orbital-stability condition. In particular, the synthetic example has $E_G/E_3 = 1/0.3$, outside that small-ratio assumption. Substituting it into the restricted expression would even give $r_H > D$. That extrapolation is not valid. The interval 4.735314–6.078802 at $D = 20$ is justified by Eqs. (11)–(12), not by a Hill-radius argument.

The $D = 1.25$ initial state likewise offers no strong separation of inner and outer scales. Its computed close approach is not surprising in that geometry, but no quantitative hierarchical-stability criterion was evaluated here. A close-approach stop is not an established collision or a proof of a particular instability mechanism. Stability criteria require their own initial-condition and assumption checks in addition to the instantaneous field-error bounds.

6.3 The equilateral benchmark and linear instability

The circular equilateral configuration is the classical Lagrange solution. The Gascheau–Routh linear criterion, expressed for the positive coefficients of Eq. (1), is [15]

$$\beta = \frac{27(E_1E_2 + E_2E_3 + E_3E_1)}{(E_1 + E_2 + E_3)^2} < 1. \tag{14}$$

For the equal case $E_i = 1/3$, $\beta = 9 > 1$, so the circular equilateral solution is linearly unstable. This concerns perturbations of that particular solution, not all trajectories of three equal sources. An exactly prescribed solution remains an exact solution. A finite-precision integration can excite growing perturbations. Its short analytic comparison and its longer departure must therefore be reported separately. Demonstrating the perturbation growth rate would require a dedicated sensitivity experiment, which is not claimed here.

7 Published systems: comparisons at three distinct levels

7.1 Kepler-16: binary and circumbinary planet

The published solution and double-lined radial velocities [11, 12] provide a test beyond synthetic trajectories. For six paired measurements and fixed published $q = 0.2937$, the binary relation $RV_A + qRV_B = C$ gives $\chi^2 = 4.248734$ for five degrees of freedom after fitting one constant. Including the small model planetary contribution changes this to 4.33655. The result tests this velocity relation, not the whole planetary solution.

A separate three-body propagation from rounded published epoch parameters, profiling only component velocity offsets, gives component RMS residuals 0.059107 and 0.878908 km s^{-1}. The first component has $\chi^2 = 32.013$ for five degrees of freedom, so a uniformly precise observational match is not obtained. Halving the 0.02-day step changes predicted velocities by only 1.45×10^{-8} km s^{-1}. Numerical step error cannot explain that residual.

Across 917 planetary epochs over 228.776 days, at radii 0.69463–0.71139 AU, the monopole field error is 1.537–9.906%. Retaining the quadrupole reduces it to 0.259–2.931%. Both compare approximations of the adopted stellar field, not measurements of a medium or newly measured planetary positions.

7.2 KOI-126: an inner binary with an outer star

KOI-126 [13] provides a compact hierarchy with inner and outer periods 1.76713 and 33.9214 days. The 16-epoch radial-velocity comparison over 436.859 days has RMS 0.209441 km s^{-1} and a maximum residual 0.612134 km s^{-1}. The numerical comparison changes velocities by at most 0.00005025 km s^{-1}. These residuals require explanation before claiming an observational reproduction.

Over 1361 sampled states in an outer period, necessary directional bounds exclude a simultaneous 5% pair-only region at those states. The best lower bound varies from 7.976% to 18.946%. This illustrates why an observed hierarchical system need not possess a clean, isolated-pair spherical region at an arbitrary tolerance. At a common exterior radius of 1.464941 AU for all three stars, the monopole and quadrupole errors are respectively 0.862–3.308% and 0.070–0.531%.

7.3 TIC 454140642: a $2 + 2$ quadruple

The observed quadruple [14] permits a differential test without treating four separated stars as one local centre. Let $G = (a, b)$ be the receiving pair, H the other pair, $\mathbf{r} = \mathbf{r}_b - \mathbf{r}_a$, and $\mathbf{D} = \mathbf{c}_H - \mathbf{c}_G = D\mathbf{n}$. Define

$$\mathbf{T}_{\text{exact}} = \mathbf{g}_H(\mathbf{r}_b) - \mathbf{g}_H(\mathbf{r}_a), \tag{15}$$

$$\mathbf{T}_{\text{mono}} = E_H \left[\frac{\mathbf{c}_H - \mathbf{r}_b}{|\mathbf{c}_H - \mathbf{r}_b|^3} - \frac{\mathbf{c}_H - \mathbf{r}_a}{|\mathbf{c}_H - \mathbf{r}_a|^3} \right], \tag{16}$$

$$\mathbf{T}_{\text{lin}} = \frac{E_H}{D^3}[3\mathbf{n}(\mathbf{n} \cdot \mathbf{r}) - \mathbf{r}]. \tag{17}$$

Then $\ddot{\mathbf{r}} = -E_G\mathbf{r}/r^3 + \mathbf{T}_{\text{exact}}$ exactly under Eq. (1). The leading fractional tidal scale is

$$\frac{|\mathbf{T}_{\text{lin}}|}{E_G/r^2} = \frac{E_H}{E_G} \left(\frac{r}{D}\right)^3 \sqrt{1 + 3\cos^2\theta}. \tag{18}$$

The cube arises from a spatial gradient and comparison with the inner acceleration. It is not an identity for a radial velocity invariant $u_r^2 R^3$.

Table 3: Largest sampled tidal approximation errors in the quadruple.

Error, normalized to $\lvert\mathbf{T}_{\mathrm{exact}}\rvert$	Pair A	Pair B
Collapse of external pair to its centre	1.573951%	2.557152%
Subsequent linearization	1.187377%	0.712683%
Combined vector error	2.594824%	2.580211%
Combined error / internal E_G/r^2	0.010277%	0.005206%

The calculation uses 3457 frozen model states over 432.1 days, one published osculating outer-period interval, not an exactly closed orbit. Centre separation ranges from 1.254029 to 2.459393 AU. Halving the temporal sampling changes the reported maxima by 0.008393 and 0.003408 percentage points. A 50-digit calculation at 48 epochs agrees to 4.26×10^{-19} AU day^{-2}. Collapse and linearization errors are vectors and can cancel. Their norms must not be added and called the actual combined error.

For the seven late spectra, a separate offset-cancelling differential-RV check, including the implemented light-travel correction, retains RMS residuals of 5.426 and 4.434 km s^{-1} for the two pairs. Their origin remains unresolved. The tidal percentages in the table therefore cannot be presented as the accuracy of the observed four-star motion. Published photodynamical inputs and derived velocities also carry shared modelling assumptions and extraction uncertainties.

7.4 A six-star hierarchy: published-parameter RV reconstruction

TIC 168789840 contains three eclipsing binaries, with A and C forming an inner quadruple and B in a wider orbit [16]. We reconstruct the 12 published velocities of each primary from Tables 2–3 of the cited preprint, retaining all 36 points, including flagged blended measurements. No new parameters are fitted. Each γ_j is held constant: this reconstruction does not model changes of the binary centre velocity along its outer orbit or perform a full six-source propagation. Such changes are a possible omitted contribution, not an established explanation of the residuals:

$$RV_j(t) = \gamma_j - K_j \sin[2\pi(t - T_{0j})/P_j]. \tag{19}$$

The global periods are from Table 2. All descending-node epochs and RV amplitudes are from Table 3. Circular inner orbits are the source's adopted fit, not an assumption applied to all multiple-star configurations. These data contributed to the published fit: reproduction is not

Table 4: Unweighted reconstruction residuals, normalized by orbital amplitude K, not by instantaneous RV.

Primary	K [km s^{-1}]	RMS [km s^{-1}]	100 RMS/K
A1	63.70	3.61598	5.67658%
B1	44.28	0.29195	0.65932%
C1	78.90	5.59712	7.09394%

an independent prediction. Individual error covariances are unavailable, and the source scales relative errors using the fitted residuals. No confidence level is inferred here. Table 2 lists C1's RV epoch one day later than Table 3. Substituting that alternative raises its RMS to 63.23 km s^{-1}. This source inconsistency is disclosed, not silently corrected by fitting.

For a circular binary, a measurable projected combination is

$$F_1 = \frac{PK_1^3}{2\pi} = \frac{E_2^3}{(E_1 + E_2)^2} \sin^3 i. \tag{20}$$

It follows from $K_1 = (2\pi/P)a_1 \sin i$, $a_1/a = E_2/(E_1 + E_2)$, and $(2\pi/P)^2 a^3 = E_1 + E_2$ within the binary model. The values for A1, B1 and C1 are respectively 5.5803, 9.8102 and 8.8200 in 10^9 km^3 s^{-2}. They do not determine individual E_i or the total sextuple E without additional ratios and inclinations. Thus this case adds real six-star configuration evidence and an observable constraint, while leaving a full six-body trajectory and the medium mechanism unvalidated. The tabulated amplitudes and reconstruction residuals specify the comparison reported here.

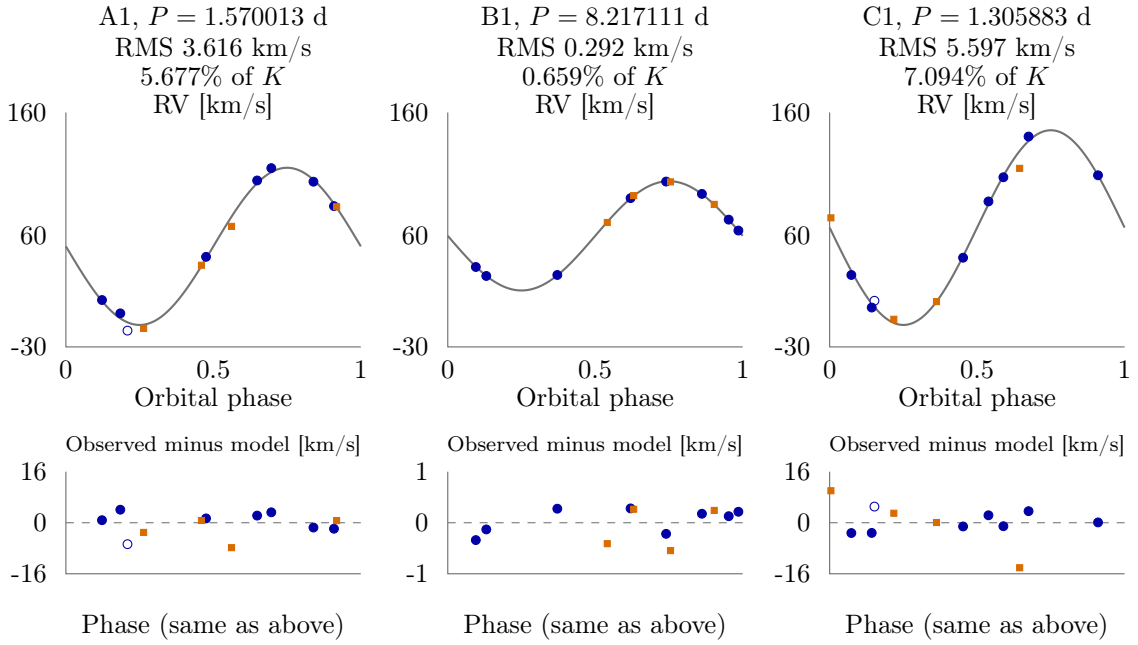


Figure 2: Published-parameter reconstruction for TIC 168789840 [16]. Solid curves: fixed tabulated circular RV models. Circles: CHIRON. Squares: TRES. Open circles: source-flagged blends. Lower panels have explicitly different residual scales. Individual observational uncertainties are not supplied as independent errors, so no invented error bars are drawn.

7.5 A predeclared outer-to-inner period diagnostic

As a bounded geometric-pattern diagnostic, we fix the observable $Q_j = T_{\rm outer}/T_j$ before evaluation and retain both inner pairs of the TIC 454140642 2+2 input parameters. This record is attributed to the osculating parameters of Kostov et al. [14] at BJD 2458437.5. It supplies $T_A = 13.61594$ d, $T_B = 10.39063$ d and $T_{\rm outer} = 432.1$ d, giving

$$Q_A = 31.7348636965, \qquad Q_B = 41.5855438987. \tag{21}$$

The displayed digits support arithmetic reproduction, not an observational precision claim. These are the two nested hierarchies in one system, not two independent systems or the ratio between sibling binaries. Both share the outer-period input and the published model. A subsequent direct check of Table 8, its notes and the period discussion on page 15 of Kostov et al. [14] supplies the uncertainties not included in the adopted input. The period row gives $T_A = 13.61594 \pm 0.00016$ d, $T_B = 10.39063 \pm 0.00008$ d and $T_{\rm outer} = 432.1^{+0.5}_{-0.4}$ d. The source distinguishes these instantaneous osculating anomalistic parameters from observed eclipse-spacing periods. No joint period covariance is supplied in the table, and a ratio of marginal medians need not be a posterior median for the ratio.

The periods are retained as model-input osculating quantities, not inferred from a new integration or replaced by instantaneous angular rates on circular orbits. No powers, Fibonacci

indices or favourable subsets were selected. Fifty-digit decimal division agrees with floating-point evaluation to within 1.14×10^{-15} absolute. This verifies arithmetic only. The adopted periods, both ratios and the sensitivity calculation are specified below. The unavailable joint period covariance remains a limitation of this comparison.

For a bounded sensitivity check using those quoted marginal endpoints, define

$$Q_j^- = \frac{T_{\text{outer}} - \Delta T_{\text{outer}}^-}{T_j + \Delta T_j^+}, \qquad Q_j^+ = \frac{T_{\text{outer}} + \Delta T_{\text{outer}}^+}{T_j - \Delta T_j^-}. \tag{22}$$

These are extrema over a rectangle of marginal endpoints, not a joint confidence interval or bounds on all systematic uncertainty. They give $[31.705114, 31.771959]$ for A and $[41.546728, 41.633985]$ for B. The maximum fractional endpoint shift is below 0.117%.

For the three predeclared direct equalities $Q = \varphi$, $Q = 3/2$ and $Q = 2$, with $\varphi = (1 + \sqrt{5})/2$, define the descriptive difference $D_c = 100|Q/c - 1|$. All six comparisons are retained:

Pair	D_φ	$D_{3/2}$	D_2
A	1861.322%	2015.658%	1486.743%
B	2470.128%	2672.370%	1979.277%

These percentages measure separation from the constants, not observational error. Even the lowest endpoint exceeds the largest tested constant. Thus this fixed direct-ratio test does not support any of the three equalities for this one modelled system. No Gaussian significance, general absence of patterns or rejection of the wider research programme follows. The comparison retains both inner pairs and all three predeclared constants.

8 Conclusions

Source parameters. We characterize each celestial source by its own orbital parameter. For an isolated binary the relative parameter is exactly $E_1 + E_2$, and the individual E_1, E_2 follow from Eq. (7) or, for double-lined spectra, from the projected relations given above.

Controlled group approximation. We derive the explicit sufficient bound $B_G = 3m_2R^2/(R-b)^4$ for replacing a group by one centre, together with the external-companion bound, Eq. (12). For the pair-plus-third configuration at $D = 20$, the initial radial interval 4.7353–6.0788 keeps each of the two errors below 5% in every direction. Closer configurations need not admit such an interval.

Coupling between two pairs. We show that the leading differential contribution scales as $(E_H/E_G)(r/D)^3\sqrt{1+3\cos^2\theta}$, Eq. (18). In TIC 454140642 its largest sampled discrepancy against the exact prescribed differential acceleration is 2.595%.

Observed systems. Kepler-16, KOI-126, TIC 454140642 and the sextuple TIC 168789840 are calculated through orbital parameters, with residuals stated in the corresponding sections. Numerical accuracy, approximation error and observational residuals are reported as separate quantities.

Connection to the three- and N-body problem. For positive sources the equations have the form studied in celestial mechanics. Established coordinates, multipole methods and stability criteria therefore apply in orbital parameters, as shown for the Lagrange equilateral benchmark.

Next step. Long-term evolution, full refits of the observed systems and predeclared period tests for further multiple systems extend these results. The direct period equalities tested in TIC 454140642 are not supported, and this negative result is retained.

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AI-based tools were used to assist with calculations, code development and manuscript preparation. The author takes full responsibility for the final content. No new experiment or observation is reported here.

Data and code availability

An earlier Ew2R3 visualization release is archived on Zenodo [7]. Calculations, input data and interactive visualization for this study are part of the Ew2R3 software release [8]. The programme website is Ew2R3.org.

Published orbital parameters and radial velocities are taken from the sources cited in the corresponding sections. Synthetic trajectories are outputs of the numerical procedure described here and are distinguished from observations.

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A Proof and normalization of the sufficient bounds

For $\mathbf{F}(\mathbf{x}) = \mathbf{x}/|\mathbf{x}|^3$, $\|D\mathbf{F}\| \le 2/|\mathbf{x}|^3$. Direct differentiation gives

$$D^2\mathbf{F}[\mathbf{h},\mathbf{h}] = -\frac{6\mathbf{h}(\mathbf{x}\cdot\mathbf{h}) + 3\mathbf{x}|\mathbf{h}|^2}{R^5} + \frac{15\mathbf{x}(\mathbf{x}\cdot\mathbf{h})^2}{R^7}. \tag{23}$$

Resolving $\mathbf{h}$ parallel and perpendicular to $\mathbf{x}$ yields $\left\|D^2\mathbf{F}[\mathbf{h},\mathbf{h}]\right\| \le 6|\mathbf{h}|^2/R^4$. Taylor's integral remainder supplies the factor $1/2$. On each source-to-centre segment the distance from the field point is at least $R - b$. The weighted first-order term vanishes because $\sum E_i\mathbf{s}_i = 0$. Summing the remainders and dividing by E_G/R^2 proves Eq. (11).

For the third source, first bound the field difference between a trial point and the centre by $2E_3R/(D-R)^3$. Then apply the same second-order remainder to the weighted acceleration of the pair centre, giving $3E_3m_2/(D-b)^4$. Their sum, divided by E_G/R^2, proves Eq. (12). These are bounds on vector accelerations at one instant. Integration of the error over time is a separate problem and is not asserted here.

B Reproduction and evidence levels

The synthetic tests use the length and time units L_0, T_0 defined in the numerical protocol. Published-system calculations use astronomical units and days internally, with velocities converted to km s^{-1} for comparison with spectra. A radial-velocity comparison refers to a specified observation time, component identification, viewing direction and velocity zero point. In this study these conventions follow the cited observational inputs, and fitted offsets are identified in the corresponding comparisons.

Algebraic identities, finite-step numerical checks, approximation errors against a specified field, and residuals against observations are four distinct forms of evidence. Accuracy at one level does not establish accuracy at the next. The reported close-approach stop, the instability-sensitive triangle comparison and the observed velocity residuals therefore retain their stated limitations.